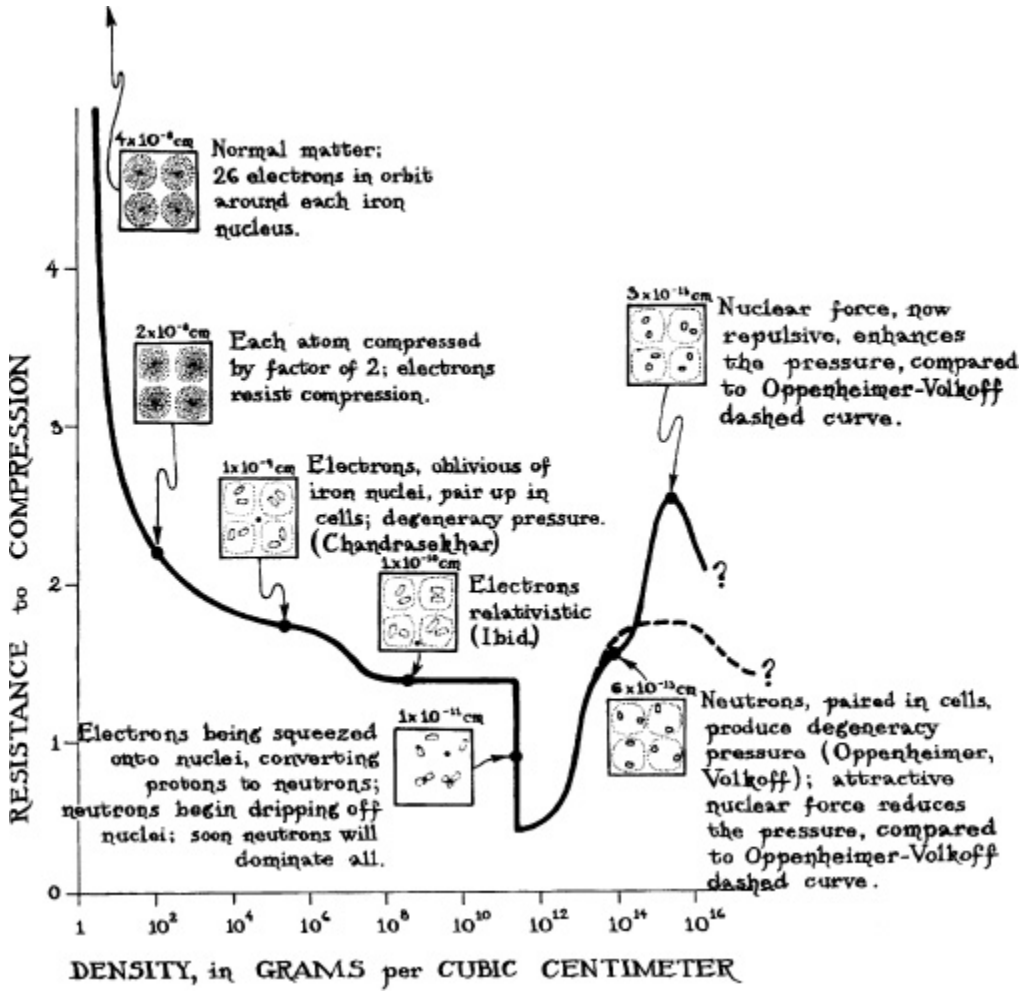
Box 5.5

The Harrison–Wheeler Equation of State for Cold, Dead Matter



The drawing above depicts the Harrison–Wheeler equation of state. Plotted horizontally is the matter's density. Plotted vertically is its resistance to compression (or adiabatic index, as physicists like to call it)—the percentage increase in pressure that accompanies a 1 percent increase in density. The boxes attached to the curve show what is happening to the matter, microscopically, as it is compressed from low densities to high. The size of each box, in centimeters, is written along the box's top.

At normal densities (left edge of the figure), cold, dead matter is composed of iron. If the matter's atomic nuclei were heavier than iron, energy could be released by splitting them apart to make iron (nuclear